What a CPT-symmetric universe adds to the Einstein equation of state

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Abstract

Jacobson derived the Einstein equation from the Clausius relation at local Rindler horizons, taking the temperature from the Unruh effect, the area law from vacuum entanglement and the horizons from local flatness. By the Bisognano-Wichmann theorem the modular conjugation of a wedge is CPT composed with a rotation by π in the transverse plane, and the same theorem makes the vacuum thermal on the wedge. Boyle, Finn and Turok take CPT to be a symmetry of the universe and not only of its laws, which folds spacetime into two sheets. Read at a horizon, their fold is the wedge conjugation composed with the transverse antipode, which is CPT itself, so Jacobson's inputs and the folded cosmology rest on one map. We re-derive his argument from that map with the metric a fixed background and say what the fold adds. Of nineteen properties of general relativity that follow from four inputs, nine were in Jacobson's argument and four are standard. The fold adds six, among them four large dimensions and a classical-quantum split with a definite weight, and one non-local term in the field equations that acts only inside black holes. Four observations would refute the construction, two of them cosmological measurements now being made.

Main

General relativity is usually treated as a classical theory awaiting quantisation. Jacobson argued in 1995 that it may not need one: the Einstein equation follows from the Clausius relation applied to the local horizons an accelerated observer sees, which makes it an equation of state [3]. He took the horizon temperature from the Unruh effect [8], the entropy proportional to area from the entanglement of vacuum fluctuations across the horizon [9,10], and horizons through every point from local flatness. Work since has obtained the field equations from horizon thermodynamics and entanglement by several routes [11-14]. One antilinear map underlies Jacobson's quantum inputs, and the same map defines a cosmology with consequences of its own.

The map is CPT, taken to hold of the universe rather than only of its laws. A law-level CPT relates one process to another. A universe-level CPT says the spacetime is folded: two sheets, and an antilinear involution Θ carrying each to the other. Boyle, Finn and Turok took this of the bang and found a cosmology with no inflaton [1,2]. This Letter applies the same assumption at a horizon.

Write $\Theta = J \circ P_\perp$, with J the antilinear modular conjugation of a wedge and $P_\perp$ the antipodal map of the transverse directions. Bisognano and Wichmann showed that J is CPT composed with a rotation by π in the transverse plane [4]. In four dimensions $P_\perp$ is that rotation, so Θ is CPT itself, with the usual $(-1)^F$ on fermions. The same theorem makes the vacuum thermal on the wedge at the Unruh temperature, so the map behind Jacobson's temperature is the map that folds the universe. Two properties carry everything that follows. First, $\Theta^2 = 1$, so the map is an involution and every field splits into the average of its two readings, Φ_c, and their difference, Φ_q. They obey the commutation algebra of the classical and quantum variables of the Keldysh formalism, in use since the 1960s as a bookkeeping device and here given a reason to exist: the classical world is where the two sheets agree. The second contour leg sits half a thermal period from the first, so unlike on the usual contour the quantum variable has a variance of its own, and its weight is fixed below. Second, $d\Theta = -\mathrm{Id}$. That is the property which survives a change of frame, and it is the one the derivation uses.

The first property needs a moment, because it is the part of the fold that a single sheet does not have. The fold keeps the divide between classical and quantum behaviour, says where it lies, and fixes its weight. For a single mode, write $W(0)$ for the correlator on one sheet at zero separation and W_{cross} for the correlator

between the sheets. Then $\langle\Phi_c^2\rangle = \frac{1}{2}[W(0) + W_{\rm cross}]$ and $\langle\Phi_q^2\rangle = 2[W(0) - W_{\rm cross}]$, and $W_{\rm cross}$ evaluated at the half-period shift derived below gives, from the KMS condition alone,

$$\frac{\langle\Phi_q^2\rangle}{\langle\Phi_c^2\rangle} = 4\tanh^2\frac{\beta\omega}{4}.$$

At frequencies below the horizon temperature the quantum half switches off as $(\beta\omega)^2/4$. That is a classical limit reached in temperature and not in $\hbar$, and it says the classical world is a horizon seen from below its own temperature. The two halves carry equal weight at $\beta\omega = 2\ln 3$. Moving the shift off a half period moves that number, so it belongs to the fold rather than to thermality.

The same construction says when the classical world switches on, and ties that moment to a particle mass: M_1, the mass of the dark-matter fermion the fold fixes at the bang [5]. Both are functionals of the same occupation, so the production integral I that fixes M_1 also fixes when the two branches stop interfering. With the mass going as $I^{-2/5}$ and that time as $M_1^{-1}I^{-2/3}$, eliminating I between them leaves

$$t_{\rm dec} \propto M_1^{2/3},$$

with $-1+\frac{5}{3} = \frac{2}{3}$ and nothing approximate in the step, so the exponent is exact and carries no free parameter. At the mass adopted in ref. 5 that time is 1.417×10^{-32} s, and 2.249×10^{-32} s if a Majorana pair is counted. The normalisation is state dependent and the power is not, which is why we quote the power.

Jacobson's argument, run on the fold. Jacobson obtains the Einstein equation by applying $\delta Q = T\,dS$ to a local Rindler horizon through every point in every null direction [3]. Because that step carries the whole of what follows, we re-derive it. Integrating Raychaudhuri's equation near the bifurcation surface, where the expansion and shear are first order so their squares are second, gives $\theta = -\lambda R_{kk}$. The heat flux carries the same $\int\lambda\,d\lambda\,dA$, and the surface gravity cancels between the two sides, which is what turns a statement about one accelerated observer into a field equation. What remains is $2\pi T_{kk} = \eta R_{kk}$ for every null k. That a symmetric tensor annihilating every null vector is a multiple of the metric is linear algebra, and we check it numerically. Built from sixty random null vectors, the constraint map has a one-dimensional kernel, its ninth singular value larger than its tenth by a factor of 3×10^{15}, and that kernel is g_{ab} to 9×10^{-16}. So $2\pi T_{ab} = \eta R_{ab} + f\,g_{ab}$ for some function f. Timelike vectors leave no kernel at all, so nullity is doing the work.

Four things go into that argument. One is the Clausius relation, the thermodynamic postulate it is built on. The other three are a temperature at every such horizon, an entropy proportional to the horizon area with a universal coefficient η, and horizons of that kind everywhere. Jacobson took the first two from the quantum field theory of the vacuum and the third from local flatness. Below, each is re-derived from Θ, which shows that they come from the same map as the cosmology. None of the three is new.

A temperature at every horizon. At a bifurcate Killing horizon the fold's map acts as a shift of Killing time by half the periodicity, so the cross-sheet correlator is the direct one at $t - i\beta/2$. That is a thermofield double at $\tanh r = e^{-\beta\omega/2}$. No temperature was put in, and none is in the Unruh effect either: by Bisognano-Wichmann the two are one statement. In modular language the half-period shift is the $\Delta^{1/2}$ of Tomita-Takesaki theory, which makes the map the square root of the thermal transformation and lets primitivity pick the fundamental period.

Horizons at every boost and orientation. In any orthonormal frame the wedge reflection, which is how J acts on coordinates, is $\mathrm{diag}(-1,-1,+1,+1)$, and the transverse antipode $P_\perp$ is $\mathrm{diag}(+1,+1,-1,-1)$. Neither factor is frame-independent. Their composition $-\mathrm{Id}$ is. A single map is therefore the fold of every Rindler wedge through its own fixed point, at every boost and every orientation. Composing the two factors in two hundred randomly boosted and rotated frames returns $-\mathrm{Id}$ to 3×10^{-14} while the reflections themselves move by order unity. Jacobson had these horizons from local Poincaré symmetry; here they come with the map.

An entropy proportional to area. If $S \propto A$ were taken from Bekenstein-Hawking it would have come out of general relativity and been fed back into a derivation of general relativity. Jacobson did not take it from

there, and neither does the fold: he motivated it from vacuum entanglement across the horizon [9,10], and that entanglement is what is computed here. Tracing out one wedge of the thermofield double that the fold requires leaves occupation $\sinh^2 r$. With $\tanh r = e^{-\beta\omega/2}$ that is exactly the Bose factor, so the entanglement entropy of a wedge is the thermal entropy of its modes, $s = (1+n)\ln(1+n) - n\ln n$. A horizon's modes carry a transverse momentum, the transverse directions are translation invariant, and the mode count of a patch is therefore proportional to its area. Every ingredient is flat-space field theory with nothing gravitational in it.

Proportionality is half of what Jacobson needs. The same η must appear at every horizon, and $\beta = 2\pi/\kappa$ differs from one horizon to the next, so the occupation visibly carries the surface gravity. It cancels. In the observer's own proper frequency the WKB mode count below ω carries κ in three places, and the Jacobian $d\omega = \kappa\, d\Omega$ cancels the $1/\kappa$ the density carries. Computed with κ kept throughout, S/A does not move to one part in 10^9 while β runs from 251 to 0.025. This follows from the theorem the fold's own map is read from: the wedge metric in boost coordinates contains no κ, and Bisognano-Wichmann fixes the modular temperature at 2π in boost time at every wedge [4], so $\beta\omega = 2\pi\Omega$ identically. Let the modular temperature depend on κ instead and S/A spreads by a factor of 21.6.

One condition attaches, and we found it by trying to break the result. S/A depends on the cutoff as $1/\epsilon^2$, so a cutoff allowed to track the surface gravity as $1/\kappa$ moves the answer by a factor of 52. The coefficient is the same at every horizon provided the ultraviolet cutoff is one proper length and not one per horizon, which is what Jacobson's single fundamental length already assumed. Whether that cutoff is finite and the same for every species is the input his later work names as still open [11], and it stays open here. The mode count itself imposes no condition. Weighting the transverse density by $e^{-k\epsilon}$ or by $1/(1+k^2\epsilon^2)$ moves S/A from 2.137 to 1.629 and 1.931 while leaving the κ independence at four parts in 10^{10}, so the cancellation belongs to the boost structure and not to one prescription.

What is claimed here is weaker than induced gravity [15,16]. Induced gravity computes $1/G$ from a given matter content and cutoff, and stands or falls on whether the species count comes out. We claim nothing of that kind. The argument needs only that the coefficient be the same at every horizon, and takes its value from experiment, which is what $8\pi G = 2\pi/\eta$ then reads as. Because η is fixed by a modular temperature and a transverse mode count, and neither is cosmological, Newton's constant cannot run, as it cannot with Jacobson's single fundamental length.

Horizons away from the fold's fixed points. The fold's own fixed points are isolated, so at a generic point no global fold acts. What the balance needs there is weaker than a global symmetry: a map whose differential is $-\mathrm{Id}$, and one that is an isometry to the order the balance is computed at. The geodesic symmetry $\exp_p v \mapsto \exp_p(-v)$ supplies both at every point of every spacetime. In normal coordinates the metric's quadratic term is built from R and its cubic term from ∇R, so the symmetry is exact through second order and fails at third. Cartan's theorem says the failure is real wherever $\nabla R \neq 0$, so what has to be settled is how heavy it is. Measured on a surface with $\nabla K \neq 0$, the residual $G_{\mu\nu}(v) - G_{\mu\nu}(-v)$ has exponent 2.9994 against 3 with a coefficient matching the closed form to five figures; on a surface whose cubic part is harmonic, so that the curvature varies while ∇K vanishes at the point, the exponent moves to 5.04. Only ∇R at the point drives the residual. Divided by the second-order term the balance uses, the residual is 1.4435 times the patch size, and vanishes with it. The field equation is that limit, which is the locally flat patch Jacobson worked in.

Parity gives a second reason the balance cannot feel the failure. A map with differential $-\mathrm{Id}$ multiplies a rank-n tensor by $(-1)^n$, and the balance contains R_{kk}, T_{kk} and a transverse area element, every one of them even. An odd-rank quantity would change sign, and the balance contains none.

What comes out. Running the balance on the fold's own horizons gives

$$G_{ab} + \Lambda g_{ab} = 8\pi G \left(T^{\mathrm{matter}}_{ab} + T^{\mathrm{img}}_{ab}[g, \Theta] \right),$$

with Λ an integration constant, where Jacobson's equation (6) also put it. The divergence step leaves $f + \eta R/2$ constant across the whole spacetime, and nothing local fixes an integration constant; a boundary condition does. Newton's constant gets its sign in the same step, since $8\pi G = 2\pi/\eta$ and η is an entropy per

unit area, built from mode entropies that are positive at every frequency, as Jacobson's $G = 1/4\hbar\eta$ already implied.

The second source is the one new term, and it makes this a different theory from general relativity with a shifted coupling. A local curvature polynomial would renormalise G and do nothing else. The image stress is not local: it depends on the world function between a point and its fold image, which is a bi-scalar. On the Einstein static universe, where homogeneity makes every local invariant constant, it runs with the separation to the image. The best local fit leaves a residual of 139 per cent of itself, while the same fitter reproduces a local stress exactly. No free parameter enters the image stress. Outside a horizon it is suppressed by mechanisms rather than by a theorem: a point and its image are spacelike separated there, so a massive field's image correlator carries e^{-md}, with md above 10^{16} for an electron at a stellar horizon. The largest value the term takes anywhere outside a horizon is 3×10^{-34} of the dark energy. It acts only inside a black hole, in a shell 2.1 microns thick at a solar mass, which is the one region where general relativity already predicts its own breakdown.

Inputs and outputs. Figure 1 counts what the fold returns against what it is given, and says where each output stood before. Nineteen properties of general relativity come out and four go in. Nine of the nineteen were in Jacobson's argument and four are standard results; the fold re-derives them, and what is new in them is their source. Six need the folded universe: the transverse involution at a black hole, the contact surfaces Raychaudhuri acts on, four large dimensions, the classical-quantum split, the sheets' opposite time orientation, and a hot bang. Where one property ends and the next begins is partly a choice, and the table in Appendix A is there so the count can be taken apart.

Two of the four inputs are what writing a metric theory means, a Lorentzian signature and one metric carrying both the horizons and the matter action. The other two are the measured constants G and Λ. Those two could not have come out, and the reason is dimensional rather than a matter of effort: in $\hbar = c = 1$ every structural input carries mass dimension zero, and $\eta = 1/4G$ has dimension two. A symmetry of dimension zero cannot produce a scale. It can produce only dimensionless numbers, and it produces eight: $w_0 = -1$ and $w_a = 0$ exactly, a two-body line at exactly half the dark-matter mass, the exponent $2/3$ tying a decoherence time to that mass, the half-period ratio that makes a horizon thermal, the equal-weight frequency $2\ln 3$, four large dimensions and no others, and the sign of G.

What would break it. The fold can be tested in ways a reformulation of general relativity cannot, because the fold that returns the field equations also fixes matter. Its state at the bang sits half in each branch of an avoided crossing, which floors how many particles are made. The observed abundance turns that floor into a ceiling, and the dark-matter fermion weighs at most 491.6 ± 2.0 PeV, with its two-body neutrino line at 245.8 ± 1.0 PeV [5]. The same construction floors the neutrino-mass sum at 58.8 meV and commits to constant dark energy, which shuts the $w_0 w_a$ escape that relaxes the present cosmological bound to 163 meV.

Four observations would end it, and none needs a threshold to be argued over: a securely measured variation in Newton's constant at any level, since there is no dial to absorb it; a robust detection of evolving dark energy, since an integration constant cannot evolve; a cosmological neutrino-mass sum below 58.8 meV; and an identified decay above the two-body endpoint. The first two test the structural half and the second two the matter sector that comes with it.

What this changes. The field equations here follow from Jacobson's argument run on the fold, and his two quantum inputs now share a source with the cosmology and its dark-matter state. They come out as general relativity plus one non-local term, which carries no free parameter and which no observation yet made could have reached. The cosmological constant is relocated rather than derived. It is a fact about the fold's global state and not about the local physics. The two places it could have come from are both closed: the turning available in the late universe is short at every finite radius, and at the bang an image stress independent of the scale factor would be of order M_1^4, some 81.4 orders above the observed value. Naming it that way is more use than calling it impossible, because it says where to look.

We do not claim that quantum gravity is impossible, or that the graviton has no commutator. The free transverse-traceless algebra keeps its own, and the ordinary weak-field mediation calculation remains available to anyone who wants it. That the field equations need not wait on a quantum theory of gravity is Jacobson's point [3]. What the fold adds is that the symmetry behind his inputs is the one that folds the universe, and

that this symmetry has consequences a single sheet lacks: four dimensions, a classical-quantum split with a definite weight, a non-local term inside black holes, and a dark-matter line that one neutrino can break.

Figure

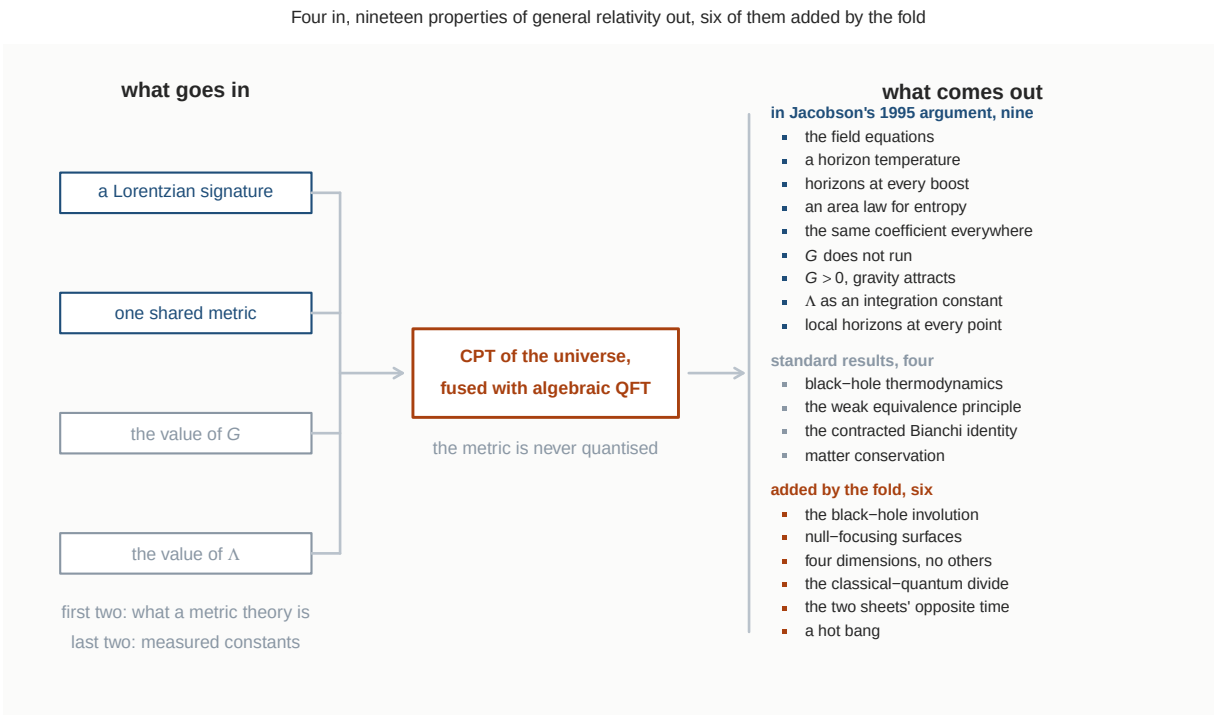


FIG. 1. Inputs and outputs. On the left is everything the fold is given; on the right is everything it returns, grouped by where each line stood before: nine in Jacobson's 1995 argument, four standard results, and six that need the folded universe. Two of the inputs are what writing a metric theory means and two are the measured constants. Full wording of every line is in Appendix A.

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The author is responsible for every claim and decision here. Literature search, grammar, proofreading, and automated tests, cross-checks, code revisions, and sanity checks used contemporary research tools, large language models among them. No such tool is an author. Appendix C names where the scripts behind these numbers live, and each check over them was validated by planting an error it had to catch, since a check that matches nothing reports success.

Competing interests

The author declares no competing interests. Linnet Labs is a company the author founded; it neither funds nor directs this work, and there is no external funding.

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Appendices

Appendix A. Inputs and outputs, line by line

Each derived line names the mechanism that produces it and, for the thirteen that were known before, where they were. The lines come from a script, and a checking pass fails if this table and the script disagree, including on which group a line sits in. Fuller wording, and the file behind each number, are in Appendix E of ref. 5.

Nineteen that come out.

Already in Jacobson's 1995 argument.

No.	What comes out	From
1	The form of the field equations	Jacobson's equation (6) [3]; here Clausius on the fold's own horizons, Raychaudhuri integrated, the null-vector algebra checked, the Bianchi step written out
2	A temperature at every local Rindler horizon	Jacobson took the Unruh effect [8]; here the map is the half-period shift, so the cross-sheet correlator is a thermofield double
3	Those horizons, at every boost and orientation	Jacobson had local Poincaré symmetry; here the wedge reflection and the transverse antipode are each frame-dependent and their composition is not

No.	What comes out	From
4	An entropy proportional to horizon area	Jacobson took vacuum entanglement with a cutoff [9]; here the entanglement entropy of the state the fold requires, with flat-space transverse mode counting
5	The same entropy coefficient at every horizon, which makes the result Einstein's rather than one equation of state per observer	Jacobson's single cutoff length; here the surface gravity cancels between occupation and WKB mode density, because Bisognano-Wichmann fixes the modular temperature at 2π
6	That Newton's constant does not run	Immediate from one coefficient; η is fixed by a modular temperature and a transverse mode count, neither of them cosmological
7	The sign of Newton's constant	$8\pi G = 2\pi/\eta$ and η is an entropy density, which is positive; Jacobson's $G = 1/4\hbar\eta$ the same
8	Λ appearing at all, as an integration constant	Jacobson's equation (6); the contracted Bianchi identity and matter conservation leave exactly one constant of integration
9	Local Rindler horizons away from the fold's fixed locus	Jacobson's local flatness; here the geodesic symmetry has differential $-\mathrm{Id}$ and is an isometry through second order, which is the order the balance uses

Standard results.

No.	What comes out	From
10	Black-hole thermodynamics	Hawking [17], Sewell from PCT [18], Kay and Wald [19]; here the same half-period shift, read at a bifurcate Killing horizon
11	The contracted Bianchi identity, which is what leaves exactly one integration constant	A theorem about the Levi-Civita connection of any metric; torsion breaks it at order unity
12	Matter conservation	Noether's second theorem on a covariant matter action, checked off shell in a generic curved metric
13	Matter following the metric's geodesics	The eikonal characteristics of a field on that metric are its geodesics, with the mass cancelling out of the path

Added by the fold.

No.	What comes out	From
14	The transverse involution at a black hole	The only free involutive isometry of a round bifurcation sphere is $-\mathrm{Id}$, and of Kerr's surface the antipodal map
15	Null-focusing surfaces for Raychaudhuri to act on	Contact is conjugacy, so the fold's contact surfaces are caustics, at every charge and dimension
16	Four large spacetime dimensions and no others	Contact needs a turning of π and the interior supplies $\pi/(D-3)$ per leg, which closes only at $D = 4$
17	The classical-quantum divide, and its amplitude	The fold's own parity; the Keldysh variables, and the symmetric thermal contour on which the weight is $4\tanh^2(\beta\omega/4)$, are standard
18	The opposite time orientation of the two sheets	The modular flow of a wedge is the boost, positive on the right static patch and negative on its antipode; the fold makes the antipode the other sheet
19	A hot bang, with no cold component at the fixed slice	Θ invariance needs a odd in conformal time, which admits a fluid only where $-3w$ is an odd integer and excludes dust

Four that go in.

No.	What goes in	Why it could not come out
1	The value of Newton's constant	Every structural input has mass dimension zero and $\eta = 1/4G$ has dimension two
2	The value of Λ	An integration constant, and both places it could have come from are closed
3	Lorentzian signature	The fold keeps a parity exactly when the number of time directions is odd, which leaves $(1,3)$ and $(3,1)$ and excludes Euclidean
4	One metric, carrying both the horizons and the matter action	T_{ab} means the variation of the matter action with respect to that metric and has no other meaning

Appendix B. Two objections to the 1995 argument

Eling, Guedens and Jacobson showed the equilibrium Clausius relation fails once the entropy is not proportional to area with a universal coefficient, and that an entropy-production term is needed in its place; their worked case is $f(R)$ [6]. Here η is not available to choose, any more than it is in Jacobson's argument with its single cutoff. It comes out of the transverse mode count with the surface gravity cancelling, so the derivation sits in the case the equilibrium relation was written for rather than assuming it does.

Chirco and Liberati showed the shear supplies an internal production term of its own, which they identify with tidal heating [7]. That term is second order at the bifurcation surface, where the expansion and the shear are both first order, and the Raychaudhuri measurement says how far out that survives: the linear behaviour holds to 4×10^{-4} at $\lambda = 0.05$ and visibly worse beyond. What remains assumed is the equilibrium reading itself, taken near the bifurcation surface of each local wedge. That reading is the Clausius relation, the first of the four things the argument uses, and the one the fold was never going to supply.

Appendix C. Code and data

Every number quoted here is produced by a script in the release accompanying ref. 5, at https://github.com/BenWiseman/sepa ways, each named for the claim it supports. A checking pass fails if a quoted number and the script that produces it come apart.